SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

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Summary table

Framework ID	Status	Fitness	Generation	Parent	Name
`fw_b9e7d103d0`	ELABORATING	0.000	0	-	Ergodic Circuit Framework
`fw_28b4bfb95f`	ELABORATING	0.000	0	-	TROPICAL_FOURIE

Details

Ergodic Circuit Framework (`fw_b9e7d103d0`)

- **Status**: `ELABORATING`
- **Fitness**: 0.000
- **Taxonomy**: COMM_COMPLEXITY
- **Target invariant**: communication_entropy_barrier (real number ≥ 0) → bounds The minimum communication complexity in a k-party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove ω(log n) lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

Primitives: - `measurable_dynamical_circuit` (tuple (C, X, μ, T)): A Boolean circuit C where gates are labeled by operations over a probability space (X, μ), and T: X → X is a measure-preserving transformation represe - `orbit_signature` (function: N → [0,1]): For a fixed input x ∈ {0,1}^n, the orbit signature is the sequence (μ(C(T^k(x))))*k, capturing how the circuit's output evolves under repeated applica - `cross_correlation_flow` (matrix ∈ R^{d×d}): For a depth-d circuit, this matrix F*{i,j} = ∫ |C_i(x) - C_i(T^{j(x))|}2 dμ(x) measures how perturbations via T propagate through layers. Computable vi

Tentative axioms: - A1: Any circuit with o(n) communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to AC^0 circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high cross_correlation_flow norm, then its communication complexity is Ω(n^δ) for some δ > 0.

TROPICAL_FOURIER_ANALYSIS (`fw_28b4bfb95f`)

- **Status**: `ELABORATING`
- **Fitness**: 0.000
- **Taxonomy**: FOURIER_ANALYTIC
- **Target invariant**: MinimalFourierCoefficient (RealNumber) → bounds The discrepancy of a function under tropical Fourier analysis.

Primitives: - `TropicalPolynomial` (Function): A function defined over a tropical semiring (max-plus or min-plus) with computable coefficients. - `TropicalFourierTransform` (Transformation): A mapping from tropical polynomials to their Fourier coefficients over a tropical semiring. - `DiscrepancyMeasure` (Metric): A function quantifying the deviation of a tropical polynomial from uniformity.

Tentative axioms: - A1: TropicalConvolution preserves the tropical semiring structure under composition. - A2: TropicalFourierTransform is invertible when restricted to certain tropical polynomials. - A3: DiscrepancyMeasure is bounded by the maximum absolute value of Fourier coefficients.